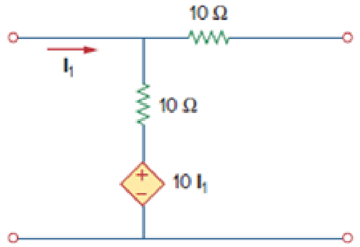
Chapter 19, Problem 6P

Problem

Compute the *z* parameters of the circuit in Fig. 19.70.

Figure 19.70



Step-by-step solution

Step 1 of 8

Refer to Figure 19.70 in the text book.

Connect the voltage source V_1 as shown in Figure 1. Identify the currents and voltages in the circuit.

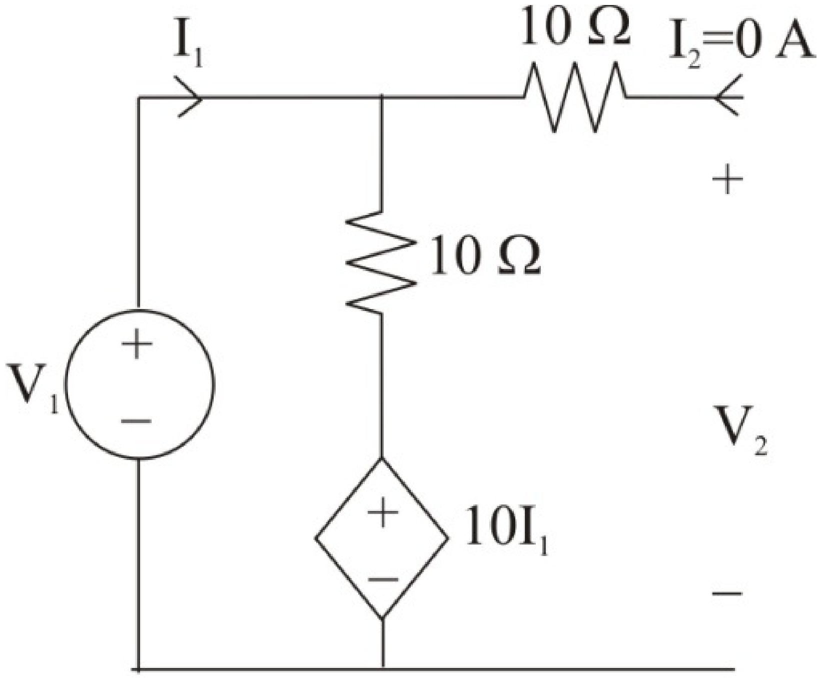


Figure 1

Step 2 of 8

Consider the expression for the voltage $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{V}_1 &= \mathbf{I}_1(10)+10\mathbf{I}_1 \\ &= \mathbf{I}_1(20)\end{aligned}$$

Therefore, the expression for the voltage $\mathbf{V}_1$ is $\mathbf{I}_1(20)$.

Consider the expression for the voltage $\mathbf{V}_2$.

$$\begin{aligned}\mathbf{V}_2 &= \mathbf{I}_1(10)+10\mathbf{I}_1 \\ &= \mathbf{I}_1(20)\end{aligned}$$

Therefore, the expression for the voltage $\mathbf{V}_2$ is $\mathbf{I}_1(20)$.

Step 3 of 8

Determine the expression for the open circuit input impedance $\mathbf{z}_{11}$.

$$\mathbf{z}_{11} = \frac{\mathbf{V}_1}{\mathbf{I}_1}$$

Substitute $\mathbf{I}_1(20)$ for $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{z}_{11} &= \frac{\mathbf{I}_1(20)}{\mathbf{I}_1} \\ &= 20\ \Omega\end{aligned}$$

Therefore, the expression for the open circuit input impedance $\mathbf{z}_{11}$ is $\boxed{20\ \Omega}$.

Step 4 of 8

Determine the value of the open circuit transfer impedance $\mathbf{z}_{21}$ from port-2 to port-1.

$$\mathbf{z}_{21} = \frac{\mathbf{V}_2}{\mathbf{I}_1}$$

Substitute $\mathbf{I}_1(20)$ for $\mathbf{V}_2$.

$$\begin{aligned}\mathbf{z}_{21} &= \frac{\mathbf{I}_1(20)}{\mathbf{I}_1} \\ &= 20\ \Omega\end{aligned}$$

Therefore, the value of the open circuit transfer impedance $\mathbf{z}_{21}$ from port-2 to port-1 is $\boxed{20\ \Omega}$.

Step 5 of 8

Connect the voltage source $\mathbf{V}_2$ as shown in Figure 2. Identify the currents and voltages in the circuit.

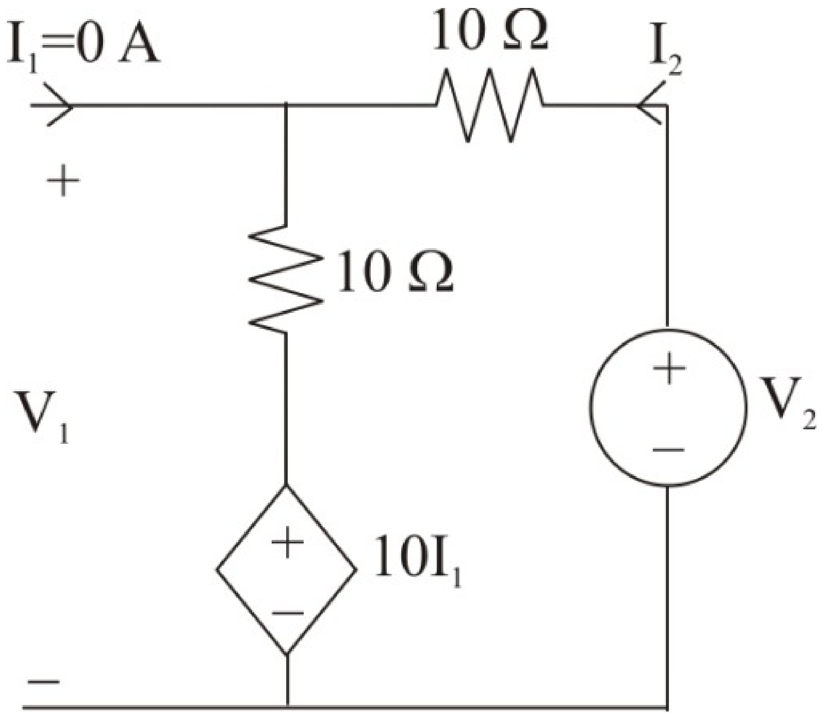


Figure 1

Step 6 of 8

Notice that the dependent source is shorted since the value of $\mathbf{I}_1$.

Consider the expression for the voltage $\mathbf{V}_2$.

$$\begin{aligned}\mathbf{V}_2 &= \mathbf{I}_2(10+10)\\ &= \mathbf{I}_2(20)\end{aligned}$$

Therefore, the expression for the voltage $\mathbf{V}_2$ is $\mathbf{I}_2(20)$.

Consider the expression for the voltage $\mathbf{V}_1$.

$$\mathbf{V}_1 = \mathbf{I}_2(10)$$

Therefore, the expression for the voltage $\mathbf{V}_1$ is $\mathbf{I}_2(10)$.

Step 7 of 8

Determine the value of the open circuit transfer impedance $\mathbf{z}_{12}$ from port-1 to port-2.

$$\mathbf{z}_{12} = \frac{\mathbf{V}_1}{\mathbf{I}_2}$$

Substitute $\mathbf{I}_2(10)$ for $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{z}_{12} &= \frac{\mathbf{I}_2(10)}{\mathbf{I}_2}\\ &= 10\ \Omega\end{aligned}$$

Therefore, the value of the open circuit transfer impedance $\mathbf{z}_{12}$ from port-1 to port-2 is $\boxed{10\ \Omega}$.

Step 8 of 8

Determine the expression for the open circuit output impedance $\mathbf{z}_{22}$.

$$\mathbf{z}_{22} = \frac{\mathbf{V}_2}{\mathbf{I}_2}$$

Substitute $\mathbf{I}_2(20)$ for $\mathbf{V}_2$.

$$\begin{aligned}\mathbf{z}_{22} &= \frac{\mathbf{I}_2(20)}{\mathbf{I}_2} \\ &= 20\ \Omega\end{aligned}$$

Therefore, the expression for the open circuit output impedance $\mathbf{z}_{22}$ is $\boxed{20\ \Omega}$.